



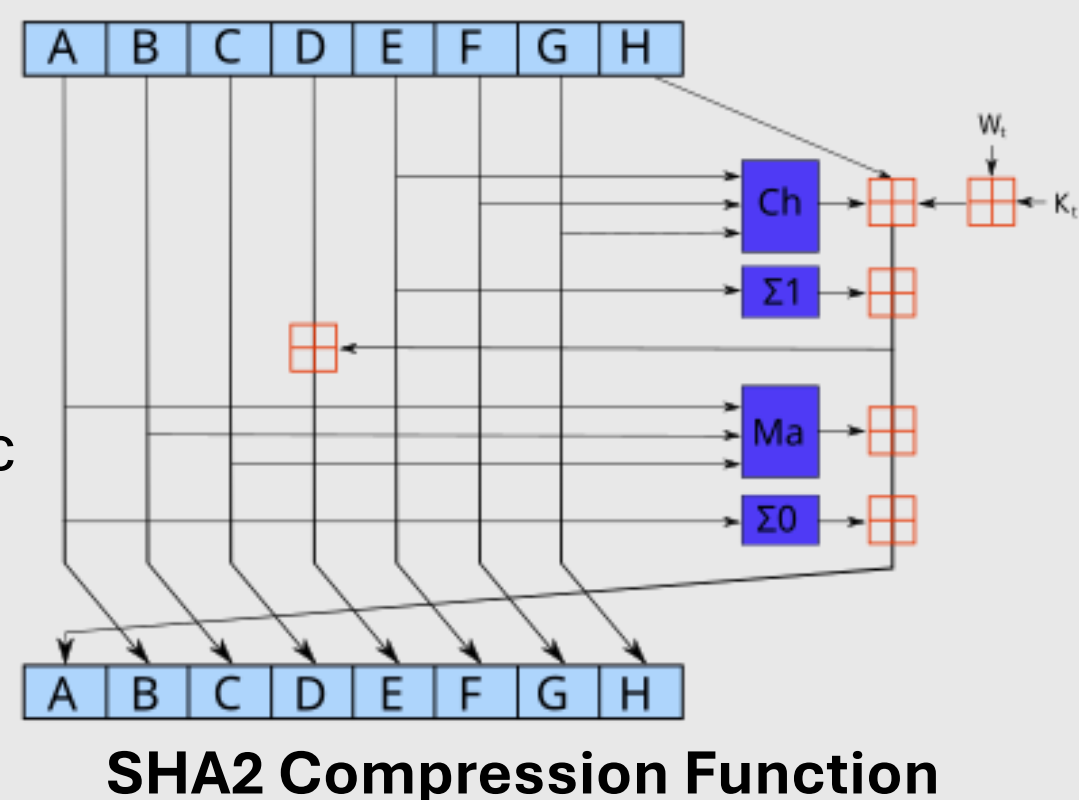
Introduction

What is SHA?

SHA-256 (Secure Hash Algorithm) is a cryptographic hash function that generates unique 256-bit outputs, widely used in security applications like blockchain, digital signatures, and data verification.

How SHA2 Works:

- **Message Preprocessing** – The input message is padded to ensure it fits into fixed 512-bit blocks.
- **Message Parsing** – The padded message is divided into 512-bit blocks for processing.
- **Initialization** – The algorithm starts with predefined hash values and constants.
- **Compression Function** – Each block undergoes multiple rounds of bitwise operations, including shifts, XORs, and modular additions, using a set of constants and functions.
- **Final Hash Output** – After all blocks are processed, the final 256-bit hash value is produced, serving as a unique digital fingerprint of the original message.



What are FPGAs?

FPGA (Field-Programmable Gate Array) is a customizable hardware chip that can be programmed to perform specific tasks, offering the flexibility of software with the speed and efficiency of dedicated hardware.

Project Summary

Summary:

This project implements SHA-256 hashing on an FPGA using a MicroBlaze soft-core processor. The Host PC sends data via UART, where the MicroBlaze processor manages communication between the SHA256 module and on-board memory. The benefits of this approach include improved speed and efficiency, as FPGA parallelism accelerates hashing compared to software-based implementations.

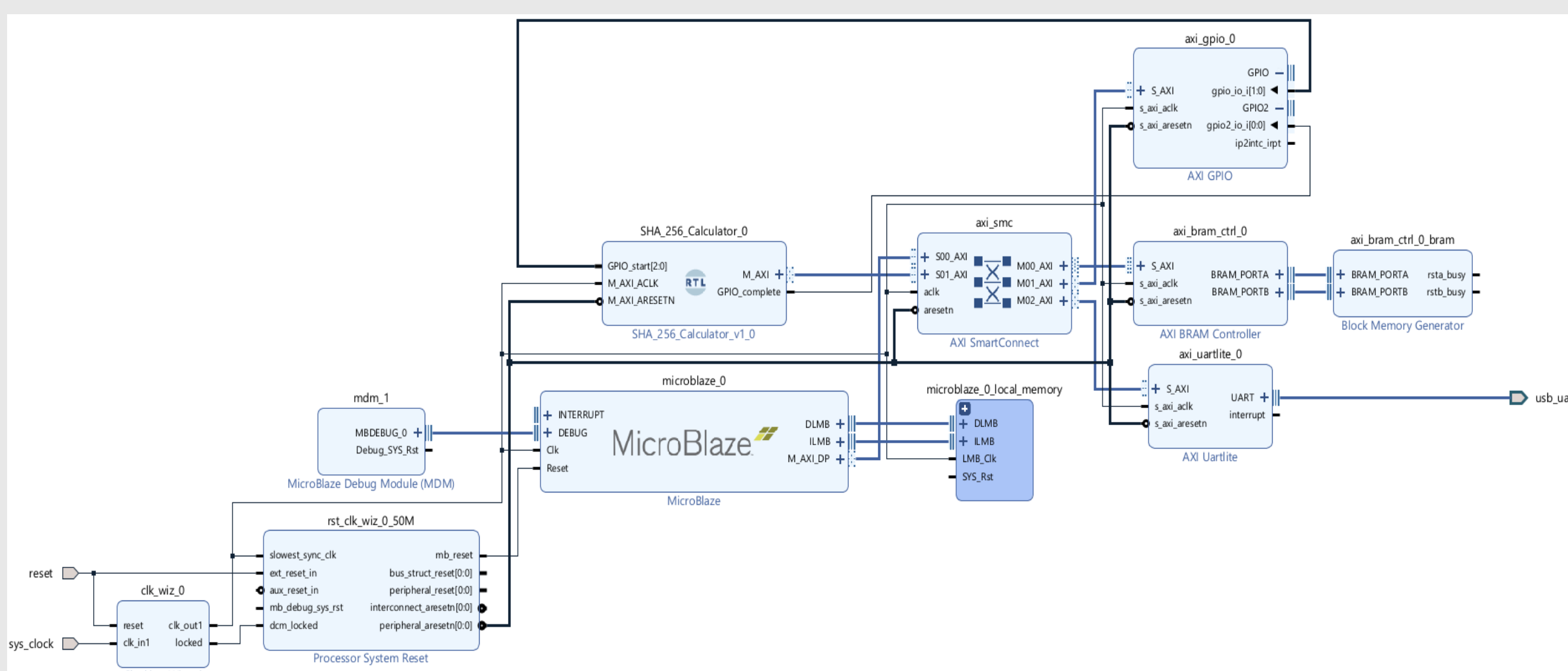
How It Works:

- **User Input:** The host PC provides data through a user interface. That gets sent to the FPGA via UART
- **MicroBlaze Microprocessor:** Microprocessor loads the message into a shared memory area for the module to process.
- **SHA256 Computation:** The SHA256 module processes the data and generates the hash. Leaving the data in a shared memory area.
- **Output Generation:** The final hash result is sent back to the host PC.

Performance Metrics:

- Achieved a clock frequency of 50 MHz
- Utilized 2,540 Look-Up Tables (LUTs), representing 12% of available resources
- Utilized 2,416 Registers, accounting for 6% of available resources
- Power Consumption: The design operates at a power usage of 0.206 Watts

Project Architecture



Key Components

- **SHA-256 Calculator:** Module for performing SHA-256 hashing computations, generating hash outputs.
- **AXI (Advanced eXtensible Interface) SmartConnect:** Hub that routes data between the MicroBlaze, SHA-256 module, and memory.
- **AXI UARTlite:** Provides a UART interface for serial communication with the host PC.
- **AXI BRAM Controller:** Manages access to Block RAM for temporary data storage during processing.

Technologies Used

Software Languages:

- **Verilog** - Hardware description language for SHA-256 implementation
- **C** - Programming for FPGA microprocessor control
- **Python** - PC-to-FPGA serial communication interface

Hardware:

- **Digilent Basys 3** - FPGA development board featuring Xilinx Artix-7 FPGA

Development Tools:

- **Vivado** - FPGA design and synthesis suite
- **Vitis** - Embedded software development platform

Design Layout

This design layout showcases the arrangement of electronic components.

Blue Areas: These represent the logic blocks, including They are responsible for executing computations and processing data.

Yellow Areas: These sections are for routing resources that facilitate communication between different components.

Green Areas: These represent memory blocks, such as Block RAM

